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BATTERY MODULE

Abstract

A battery module according to one embodiment of the present invention includes: a battery cell stack in which a plurality of battery cells are stacked; a pair of end plates provided at both ends of the battery cell stack in the stacking direction; and a cushioning material arranged either between the battery cells, or between the battery cell stack and the end plates, or both. The cushioning material includes: a first pack and a second pack with an inner diameter larger than that of the first pack; gas filled in an internal space of the first pack; and liquid filled in a space between the first pack and the second pack.

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Background/Summary

[0001] This application is based on and claims the benefit of priority from Japanese Patent Application No. 2024-035102, filed on 7 Mar. 2024, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to a battery module.

Related Art

[0003] In recent years, research and development have been conducted on battery modules that contribute to energy efficiency, in order to enable more people to access affordable, reliable, sustainable, and advanced energy. A battery module is a modularized assembly combining a plurality of battery cells, generally including a battery cell stack formed by stacking a plurality of battery cells, and a pair of end plates arranged at both ends of the battery cell stack in the stacking direction. Battery modules are used in applications requiring high current and high voltage, such as for motor drives in electric vehicles and hybrid electric vehicles.

[0004] In battery modules, consideration has been given to arranging cushioning materials either between the battery cells or between the battery cell stack and the end plates to exert pressure on the battery cells in the stacking direction. Known cushioning materials include a cushioning material having a deformable chamber and a system that supplies fluid to deform the chamber (Patent Document 1), as well as elastic bodies such as plate springs and liquid springs (Patent Document 2 and Patent Document 3). [0005] Patent Document 1: Japanese Unexamined Patent Application, Publication No. 2020-64848 [0006] Patent Document 2: Japanese Unexamined Patent Application, Publication No. 2021-96974 [0007] Patent Document 3: European Patent Application, Publication No. 3886202

SUMMARY OF THE INVENTION

[0008] In the field of battery module technology, improving the electrical capacity is a recognized challenge. An effective approach to improving the electrical capacity of a battery module is to use a cushioning material to exert uniform pressure on each battery cell incorporated into the battery module. When using a cushioning material having a function of thermal dissipation to release heat from within the battery cell stack to outside, and a thermal insulation property to limit heat transfer between the battery cells, the thermal insulation property is achieved in the cell stacking direction of the battery cell stack, and thermal conductivity is achieved in a direction perpendicular to the cell stacking direction, thereby facilitating the temperature control of the battery cells. However, in cases where the thickness of the battery cells partially fluctuates during charging and discharging, resulting in a partial increase or decrease in the pressure exerted by the battery cells on the cushioning material, it may become challenging for the cushioning material to exert uniform pressure on the battery cells. Reducing the size of a cushioning material is desirable in order to improve the electrical capacity per unit volume of the battery module; however, reducing the size of the cushioning material could compromise the balance of thermal dissipation and thermal insulation, potentially causing difficulty in achieving the intended effects.

[0009] The present invention has been made in view of the above circumstances, and aims to provide a battery module using a cushioning material that is capable of exerting uniform pressure on the battery cells, achieves high thermal insulation, and can be easily reduced in size. Consequently, the present invention contributes to increased energy efficiency.

[0010] The inventors of the present invention have found that the above problems can be solved by employing a structure of a cushioning material that includes: a double-pack structure including a first pack and a second pack with an inner diameter larger than that of the first pack, and having

both ends sealed; gas filled within the first pack; and liquid filled in the space between the first pack and the second pack, thereby arriving at completion of the present invention. Thus, the present invention provides the following. [0011] (1) A battery module, including: a battery cell stack in which a plurality of battery cells are stacked; a pair of end plates provided at both ends of the battery cell stack in a stacking direction; and a cushioning material arranged either between the battery cells, or between the battery cell stack and the end plates, or both. The cushioning material includes: a first pack and a second pack with an inner diameter larger than that of the first pack; gas filled in an internal space of the first pack; and liquid filled in a space between the first pack and the second pack.

[0012] With the battery module as described in (1), for example, even in cases where the pressure exerted by the battery cell on the cushioning material partially increases or decreases, the liquid filled in the space between the first pack and the second pack maintains the internal pressure of the cushioning material uniform, and thus the pressure exerted by the cushioning material on the battery cell is maintained uniform. Since gas is filled inside the first pack, the thermal insulation property of the cushioning material is enhanced. Furthermore, the cushioning material consists of the double-pack structure including the first pack and the second pack, which can be relatively easily configured, and easily reduced in size. [0013] (2) In the battery module as described in (1), the first pack and the second pack are spaced apart from each other.

[0014] With the battery module as described in (2), the thermal conductivity between the first pack and the second pack is reduced, thereby improving the thermal insulation performance of the cushioning material. [0015] (3) In the battery module as described in (1) or (2), the cushioning material includes an air ventilation tube connected to the internal space of the first pack.

[0016] With the battery module as described in (3), the internal pressure of the cushioning material can be regulated by introducing gas into the internal space of the first pack or exhausting the gas through the air ventilation tube. Therefore, the pressure exerted by the cushioning material on the battery cells can be maintained constant. [0017] (4) In the battery module as described in any one of (1) to (3), the cushioning material includes a liquid supply tube and a liquid discharge tube connected to the space between the first pack and the second pack.

[0018] The battery module as described in (4) is capable of regulating the internal pressure of the cushioning material by introducing liquid into the space between the first pack and the second pack through the liquid supply tube or discharging the liquid through the liquid discharge tube. Therefore, the pressure exerted by the cushioning material on the battery cells can be maintained more constant. By continuously flowing temperature-controlled liquid in the space between the first pack and the second pack via the liquid supply tube and the liquid discharge tube, the temperature of the cushioning material can be regulated. Therefore, the thermal insulation property of the cushioning material is further improved.

[0019] The present invention makes it possible to provide a battery module using a cushioning material that is capable of exerting uniform pressure on the battery cells, achieves high thermal insulation, and can be easily reduced in size.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is a cross-sectional view of a battery module according to a first embodiment of the present invention;

[0021] FIG. 2 is a cross-sectional view taken along the line II-II of FIG. 1;

[0022] FIG. 3 is a cross-sectional view of a battery cell that can be used in the battery module according to the first embodiment of the present invention;

[0023] FIG. 4 is a cross-sectional view illustrating a battery module according to a second

embodiment of the present invention;

[0024] FIG. 5 is a plan view of a cushioning material used in the battery module according to the second embodiment of the present invention;

[0025] FIG. 6 is a first modification of the cushioning material used in the battery module according to the present invention;

[0026] FIG. 7 is a second modification of the cushioning material used in the battery module according to the present invention;

[0027] FIG. 8 is a third modification of the cushioning material used in the battery module according to the present invention; and

[0028] FIG. 9 is a fourth modification of the cushioning material used in the battery module according to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0029] Hereinafter, embodiments of the present invention will be described with reference to the drawings. However, the embodiments described below are merely illustrative and do not limit the scope of the present invention.

[0030] FIG. 1 is a cross-sectional view of a battery module according to a first embodiment of the present invention. FIG. 2 is a cross-sectional view taken along the line II-II of FIG. 1. FIG. 3 is a cross-sectional view of a battery cell that can be used in the battery module according to the first embodiment of the present invention.

[0031] As illustrated in FIGS. 1 and 2, a battery module 1 includes: a battery cell stack 100 formed by stacking a plurality of battery cells 10; a pair of end plates 20 provided at both ends of the battery cell stack in the stacking direction (Z direction in FIG. 1); and a cushioning material 30 arranged between the battery cells 10 and between the battery cell stack 100 and the end plates 20. The battery cell stack 100, the end plates 20, and the cushioning material 30 are housed within a module case 40. The module case 40 includes a positive electrode terminal 51 and a negative electrode terminal 52. The positive electrode terminal 51 and the negative electrode terminal 52 are arranged in opposing positions in one direction (X direction in FIG. 1). The positive electrode terminal 51 is connected to a positive electrode lead wire 11a of the battery cell 10. The negative electrode terminal 52 is connected to a negative electrode lead wire 14a of the battery cell 10.

[0032] The battery cell 10 is a battery utilizing lithium ions as the charge transfer medium. As illustrated in FIG. 3, the battery cell 10 includes an electrode stack 18, which is formed by stacking a positive electrode layer 11 and a negative electrode layer 14 with a solid electrolyte layer 17 interposed therebetween, and a housing 19 that houses the electrode stack 18. The positive electrode layer 11 includes a positive electrode current collector 12 and a positive electrode active material layer 13. The negative electrode layer 14 includes a negative electrode current collector 15 and a metal layer 16. When the battery cell 10 is charged, lithium ions released from the positive electrode active material layer 13 pass through the solid electrolyte layer 17 and deposit on the surface of the metal layer 16 of the negative electrode layer 14, forming a lithium deposition layer that increases the thickness of the negative electrode layer 14. This lithium deposition layer acts as the negative electrode active material layer, and dissolves upon discharge by releasing lithium ions. Therefore, the volume of the battery cell 10 changes with charging and discharging. Consequently, the pressure exerted by the battery cell 10 on the cushioning material 30 changes during charging and discharging. The stacking direction of the electrode stack 18 aligns with the stacking direction of the battery cell stack 100. In other words, the plurality of battery cells 10 in the battery module 1 are stacked along the stacking direction of the electrode stack 18. In the battery cell 10 illustrated in FIG. 4, only one electrode stack 18 is housed within the housing 19, although a plurality of electrode stacks 18 may be housed within the housing 19.

[0033] The positive electrode current collector 12 is not particularly limited in material or shape so long as having a function as a current collector for the positive electrode layer 11. Examples of materials for the positive electrode current collector 12 include aluminum, aluminum alloy,

stainless steel, nickel, iron, and titanium, in which aluminum, aluminum alloy, and stainless steel are preferred. Examples of the shape of the positive electrode current collector **12** include foil and plate.

[0034] The positive electrode active material layer **13** contains at least one type of positive electrode active material. There is no particular limitation on the positive electrode active material, and may be any materials commonly used for positive electrode layers in solid-state secondary batteries. Examples of positive electrode active materials include layered active materials containing lithium, spinel-type active materials, and olivine-type active materials. Specific examples of positive electrode active materials include lithium cobalt oxide ($\text{LiCoO}_{0.5}\text{O}_{1.5}$), lithium nickel oxide ($\text{LiNiO}_{0.5}\text{O}_{1.5}$), $\text{LiNi}_{0.5}\text{pMn}_{0.5}\text{qCo}_{0.5}\text{rO}_{1.5}$ (where $p+q+r=1$), $\text{LiNi}_{0.5}\text{pAl}_{0.5}\text{qCo}_{0.5}\text{rO}_{1.5}$ (where $p+q+r=1$), lithium manganese oxide ($\text{LiMn}_{0.5}\text{O}_{1.5}$), heteroelement-substituted Li—Mn spinel such as $\text{Li}_{0.5}\text{Mn}_{1.5}\text{MO}_{4}$ (where $x+y=2$, and M is at least one element selected from Al, Mg, Co, Fe, Ni, and Zn), lithium titanate (an oxide containing Li and Ti), and lithium metal phosphate ($\text{LiMPO}_{0.5}$, where M is at least one element selected from Fe, Mn, Co, and Ni).

[0035] The positive electrode active material layer **13** may optionally contain a solid electrolyte to improve lithium ion conductivity. A conductive additive may be optionally contained to improve electrical conductivity. A binder may be optionally contained to impart flexibility or other properties. The solid electrolyte, the conductive additive, and the binder are not particularly limited, and may be any materials commonly used for positive electrode layers in solid-state secondary batteries.

[0036] The material for the positive electrode lead wire **11a** may be the same as the material for the positive electrode current collector **12**, or may be different from the material for the positive electrode current collector **12**. The positive electrode lead wire **11a** may be integrally connected to the positive electrode current collector **12**.

[0037] The negative electrode current collector **15** is not particularly limited in material or shape so long as having a function as a current collector for the negative electrode layer **14**. Examples of materials for the negative electrode current collector **15** include nickel, copper, and stainless steel. Examples of the shape of the negative electrode current collector **15** include foil and plate.

[0038] The metal layer **16** is not particularly limited in material or shape so long as having a function of densely depositing lithium ions. The metal layer **16** may be a metallic lithium layer or a layer of a metal that forms an alloy with lithium. Examples of metals that form alloys with lithium include Mg, Si, Au, Ag, In, Ge, Sn, Pb, Al, and Zn. The metal forming the metal layer **16** may be in the form of powder or a thin film. By employing the negative electrode layer **14** including the metal layer **16**, a uniform lithium deposition layer can be formed on the surface of the metal layer **16**.

[0039] The material for the negative electrode lead wire **14a** may be the same as the material for the negative electrode current collector **15**, or may be different from the material for the negative electrode current collector **15**. The negative electrode lead wire **14a** may be integrally connected to the negative electrode current collector **15**.

[0040] The solid electrolyte layer **17** contains at least one type of solid electrolyte. The solid electrolyte is not particularly limited so long as having lithium ion conductivity; examples include sulfide solid electrolytes, oxide solid electrolytes, nitride solid electrolytes, and halide solid electrolytes.

[0041] Examples of sulfide solid electrolytes include $\text{Li}_{0.5}\text{S}_{1.5}\text{P}_{0.5}\text{S}_{1.5}$ and $\text{Li}_{0.5}\text{S}_{1.5}\text{P}_{0.5}\text{S}_{1.5}\text{LiI}$. The sulfide solid electrolyte may have an argyrodite-type crystal structure.

[0042] Examples of oxide solid electrolytes include NASICON-type oxides, garnet-type oxides, and perovskite-type oxides. An example of a NASICON-type oxide is an oxide containing Li, Al, Ti, P, and O (e.g., $\text{Li}_{1.5}\text{Al}_{0.5}\text{Ti}_{1.5}(\text{PO}_4)_3$). An example of a garnet-type oxide is an oxide containing Li, La, Zr, and O (e.g., $\text{Li}_{0.7}\text{La}_{0.3}\text{Zr}_{0.2}\text{O}_{12}$). An example of a perovskite-type oxide is an oxide containing Li, La, Ti, and O (e.g., LiLaTiO_3).

[0043] The housing **19** is designed to be expandable and contractible to accommodate changes in the volume of the battery cell **10** during charging and discharging. Examples of materials that can be used for the housing **19** include a laminated film. The laminated film may be a three-layer film composed of an inner resin layer, a metal layer, and an outer resin layer, stacked in this order from the inner side. The outer resin layer may be a polyamide (nylon) layer or polyethylene terephthalate (PET) layer, the metal layer may be an aluminum layer, and the inner resin layer may be a polyethylene or polypropylene layer.

[0044] The end plates **20** function to restrain the battery cell stack **100** in the stacking direction. The restraining force of the end plates **20** allows for regulating the surface pressure exerted by the cushioning material **30** on the battery cell stack. The material for the end plate **20** is not particularly limited, and various materials commonly used for end plates for battery modules can be employed.

[0045] The cushioning material **30** functions to equalize the surface pressure exerted on the battery cells **10**. The surface pressure exerted on the battery cells **10** is, for example, 1 MPa or more.

[0046] The cushioning material **30** includes a double-pack structure **33** that includes a first pack **31** and a second pack **32** with an inner diameter larger than that of the first pack **31**. The first pack **31** and the second pack **32** are spaced apart from each other. Gas is filled in a first space **35** inside the first pack **31**. Liquid is filled in a second space **36** between the first pack **31** and the second pack **32**.

[0047] The area ratio between the first space **35** and the second space **36** in a cross-section perpendicular to the axial direction of the cushioning material **30** may be set from a standpoint of thermal insulation and thermal conductivity of the cushioning material **30**. The ratio of the cross-sectional area of the first space **35** to the total cross-sectional area of the cushioning material **30** is preferably within a range of 20 to 50% from a standpoint of thermal insulation, and is preferably within a range of 50 to 80% or less from a standpoint of thermal conductivity. Nitrogen gas, for example, may be used as the gas filled in the first space. Examples of liquids that can be used for filling the second space **36** include mineral-based hydraulic oil, phosphate ester-based hydraulic oil, water, and glycol-based solvents.

[0048] The cushioning material **30** can be manufactured by, for example, enclosing a first tubular member containing gas within a second tubular member with an inner diameter larger than that of the first tubular member, filling the space between the first tubular member and the second tubular member with liquid, and then sealing the second tubular member. Examples of materials that can be used for the first tubular member and the second tubular member include a laminated film.

[0049] The module case **40** houses the battery cell stack **100** and the end plates **20**. The material for the module case **40** is not particularly limited, and various materials commonly used for battery module cases can be employed. The material for the positive electrode terminal **51** may be the same as the material for the positive electrode lead wire **11a**, or may be different from the material for the positive electrode lead wire **11a**. The material for the negative electrode terminal **52** may be the same as the material for the negative electrode lead wire **14a**, or may be different from the material for the negative electrode lead wire **14a**.

[0050] In the battery module **1** of the present embodiment established in the configuration described above, even in cases where the volume of the battery cell **10** increases during charging, resulting in a partial increase in the pressure exerted by the battery cell **10** on the cushioning material **30**, or in cases where the volume of the battery cell **10** decreases during discharging, resulting in a partial decrease in the pressure exerted by the battery cell **10** on the cushioning material **30**, the internal pressure of the cushioning material **30** is maintained uniform by the liquid filled in the second space **36** between the first pack **31** and the second pack **32**. Consequently, the pressure exerted by the cushioning material **30** on the battery cell can be maintained uniform. The presence of the gas filled in the first space **35** inside the first pack **31** enhances the thermal insulation of the cushioning material **30**. The cushioning material **30** consists of the double-pack structure **33** including the first pack **31** and the second pack **32**, which is relatively simple in

structure, enhancing the ease of size reduction. The first pack **31** and the second pack **32** are spaced apart from each other, reducing thermal conductivity between the first pack **31** and the second pack **32**, thereby improving the thermal insulation of the cushioning material **30**.

[0051] FIG. **4** is a cross-sectional view illustrating a battery module according to a second embodiment of the present invention. FIG. **5** is a plan view illustrating a cushioning material used in the battery module of the second embodiment.

[0052] As illustrated in FIGS. **4** and **5**, the battery module **2** of the second embodiment is configured similarly to the battery module **1** described above, except that it includes an air ventilation tube **37** connected to the first space **35** of the cushioning material **30**, and a liquid supply tube **38** and a liquid discharge tube **39** connected to the second space **36**. Therefore, the same reference numbers are assigned to the components common to the battery module **1**, and descriptions thereof are omitted.

[0053] The air ventilation tube **37** functions as an inlet and outlet for the gas within the first space **35**. For example, when the volume of the battery cell **10** increases during charging, resulting in a partial increase in the pressure exerted by the battery cell **10** on the cushioning material **30** and a corresponding increase in the internal pressure of the cushioning material **30**, gas is exhausted from the first space **35** through the air ventilation tube **37**, thereby allowing for lowering the internal pressure of the cushioning material **30**. Conversely, when the volume of the battery cell **10** decreases during discharging, resulting in a partial decrease in the pressure exerted by the battery cell **10** on the cushioning material **30**, gas is introduced from the outside into the first space **35** through the air ventilation tube **37**, thereby allowing for raising the internal pressure of the cushioning material **30**. Furthermore, regardless of whether being during charging or discharging, the pressure exerted by the cushioning material **30** on the battery cell **10** can be regulated by introducing gas into the first space **35** or exhausting the gas from the first space **35** through the air ventilation tube **37**.

[0054] The liquid supply tube **38** functions as an inlet for the liquid within the second space **36**, while the liquid discharge tube **39** functions as an outlet for the liquid within the second space **36**. For example, when the volume of the battery cell **10** decreases during discharging, resulting in a partial decrease in the pressure exerted by the battery cell **10** on the cushioning material **30**, liquid is introduced from the outside into the second space **36** through the liquid supply tube **38**, thereby allowing for raising the internal pressure of the cushioning material **30**. When the volume of the battery cell **10** increases by charging, resulting in a partial increase in the pressure exerted by the battery cell **10** on the cushioning material **30** and a corresponding increase in the internal pressure of the cushioning material **30**, liquid is discharged from the second space **36** to the outside through the liquid discharge tube **39**, thereby allowing for lowering the internal pressure of the cushioning material **30**. Furthermore, regardless of whether being during charging or discharging, the pressure exerted on the battery cell **10** can be regulated by way of the cushioning material **30** to regulate the internal pressure of the first space **35**. Additionally, regardless of whether being during charging or discharging, the pressure exerted by the cushioning material **30** on the battery cell **10** can be regulated by introducing liquid into the second space **36** through the liquid supply tube **38** or discharging the liquid through the liquid discharge tube **39**. The internal temperature of the cushioning material **30** can also be regulated by supplying temperature-controlled liquid into the second space **36** through the liquid supply tube **38** and discharging liquid through the liquid discharge tube **39**.

[0055] The battery module **2** of the second embodiment achieves the same effects as the battery module **1** does, since the cushioning material **30** includes the double-pack structure **33** including the first pack **31** and the second pack **32**, gas filled in the first space **35**, and liquid filled in the second space **36**, similarly to the battery module **1**. Furthermore, the battery module **2** of the second embodiment is capable of regulating the internal pressure of the cushioning material **30** by introducing/exhausting gas into/from the first space **35** through the air ventilation tube **37**, and thus

is capable of maintaining constant pressure exerted by the cushioning material **30** on the battery cell **10**. Additionally, the battery module **2** of the second embodiment introduces liquid into the second space **36** through the liquid supply tube **38** or discharges the liquid through the liquid discharge tube **39**, thereby allowing for regulating the internal pressure of the cushioning material **30**, and thus is capable of maintaining more constant pressure exerted by the cushioning material **30** on the battery cell **10**. By continuously circulating temperature-controlled liquid in the second space **36** through the liquid supply tube **38** and the liquid discharge tube **39**, the temperature of the cushioning material **30** can be regulated, thereby further enhancing the thermal insulation of the cushioning material **30**.

[0056] In the battery module **1** of the first embodiment and the battery module **2** of the second embodiment, the cushioning material **30** includes the double-pack structure **33** including one first pack **31** inside of the second pack **32**; however, the configuration of the cushioning material **30** is not limited to this. For example, two or more first packs **31** may be provided within the second pack **32**. Modifications to the cushioning material **30** are illustrated in FIGS. **6** to **9**.

[0057] FIG. **6** is a first modification of the cushioning material used in the battery module according to the present invention; A cushioning material **30a** of the first modification includes two first packs **31a** and **31b** within the second pack **32**. The first pack **31a** and the first pack **31b** are arranged near the ends of the second pack **32**. Gas is filled in the first space **35** within the first pack **31a**, **31b**, and liquid is filled in the second space **36** between the first pack **31a**, **31b** and the second pack **32**. The gas-filled first space **35** exhibits higher thermal insulation performance, as compared to the liquid-filled second space **36**. The liquid-filled second space **36** exhibits higher cooling performance, as compared to the gas-filled first space **35**. Consequently, the cushioning material **30a** of the first modification exhibits higher thermal insulation performance at the ends of the second pack **32** and higher cooling performance at the center of the second pack **32**, as compared to the cushioning material **30**.

[0058] FIG. **7** is a second modification of the cushioning material used in the battery module according to the present invention. A cushioning material **30b** of the second modification includes an additional first pack **31c** arranged between the first pack **31a** and the first pack **31b** of the first modification. As a result, the cushioning material **30b** of the second modification achieves more uniform thermal insulation performance and cooling performance, as compared to the cushioning material **30a** of the first modification.

[0059] FIG. **8** is a third modification of the cushioning material used in the battery module according to the present invention. A cushioning material **30c** of the third modification is configured such that the first pack **31c** arranged at the center of the second pack **32** in the cushioning material **30d** of the second modification is modified to a flattened, elliptical first pack **31d**, and the second space **36** is modified to be wider. Therefore, the cushioning material **30c** of the third modification exhibits higher cooling performance at the center of the second pack **32**, as compared to the cushioning material **30b** of the second modification.

[0060] FIG. **9** is a fourth modification of the cushioning material used in the battery module according to the present invention. A cushioning material **30d** of the fourth modification is configured such that two flattened, circular first packs **31e** and **31f** are arranged at the center of the second pack **32**, in parallel along a direction orthogonal to the thickness direction of the second pack **32**. Therefore, the cushioning material **30d** of the fourth modification exhibits higher cooling performance at the ends of the second pack **32** and higher thermal insulation performance at the center of the second pack **32**, as compared to the cushioning material **30**.

[0061] As shown by the first packs **31a** to **31f** of the cushioning materials **30a** to **30d**, the balance between cooling performance and thermal insulation performance of the cushioning material can be adjusted based on the shape and arrangement of the first pack(s).

[0062] The preferred embodiments of the present invention have been described above. However, the present invention is not limited to these embodiments, and can be modified as appropriate.

[0063] For example, in the embodiments described above, the first pack **31** and the second pack **32** of the cushioning material **30** are spaced apart; however, the arrangement of the first pack **31** and the second pack **32** is not limited to this. The first pack **31** and the second pack **32** may be partially in contact with each other.

[0064] In the embodiments described above, the battery cell **10** has been described as a solid-state battery including the solid electrolyte layer **17**; however, the battery cell **10** is not limited to this configuration. The battery cell **10** may alternatively be a non-aqueous battery using an organic electrolytic solution as the electrolyte, or a polymer battery using a polymer gel (polymer).

[0065] In the embodiments described above, the cushioning materials **30**, **130** are arranged both between the battery cells **10** and between the battery cells **10** and the end plate **20**; however, the positions of the cushioning materials **30**, **130** are not limited to this arrangement. The cushioning materials **30**, **130** only need to be arranged either between the battery cells **10**, or between the battery cells **10** and the end plates **20**, or both.

EXPLANATION OF REFERENCE NUMERALS

[0066] **1, 2**: battery module [0067] **10**: battery cell [0068] **11**: positive electrode layer [0069] **11a**: positive electrode lead wire [0070] **12**: positive electrode current collector [0071] **13**: positive electrode active material layer [0072] **14**: negative electrode layer [0073] **14a**: negative electrode lead wire [0074] **15**: negative electrode current collector [0075] **16**: metal layer [0076] **17**: solid electrolyte layer [0077] **18**: electrode stack [0078] **19**: housing [0079] **20**: end plate [0080] **30, 30a, 30b, 30c, 30d**: cushioning material [0081] **31, 31a, 31b, 31c, 31d, 31e, 31f**: first pack [0082] **32**: second pack [0083] **33**: double-pack structure [0084] **35**: first space [0085] **36**: second space [0086] **37**: air ventilation tube [0087] **38**: liquid supply tube [0088] **39**: liquid discharge tube [0089] **40**: module case [0090] **51**: positive electrode terminal [0091] **52**: negative electrode terminal

Claims

1. A battery module, comprising: a battery cell stack in which a plurality of battery cells are stacked; a pair of end plates provided at both ends of the battery cell stack in a stacking direction; and a cushioning material arranged either between the battery cells, or between the battery cell stack and the end plates, or both, wherein the cushioning material includes: a first pack and a second pack with an inner diameter larger than that of the first pack; gas filled in an internal space of the first pack; and liquid filled in a space between the first pack and the second pack.
 2. The battery module according to claim 1, wherein the first pack and the second pack are spaced apart from each other.
 3. The battery module according to claim 1, wherein the cushioning material includes an air ventilation tube connected to the internal space of the first pack.
 4. The battery module according to claim 1, wherein the cushioning material includes a liquid supply tube and a liquid discharge tube connected to the space between the first pack and the second pack.
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